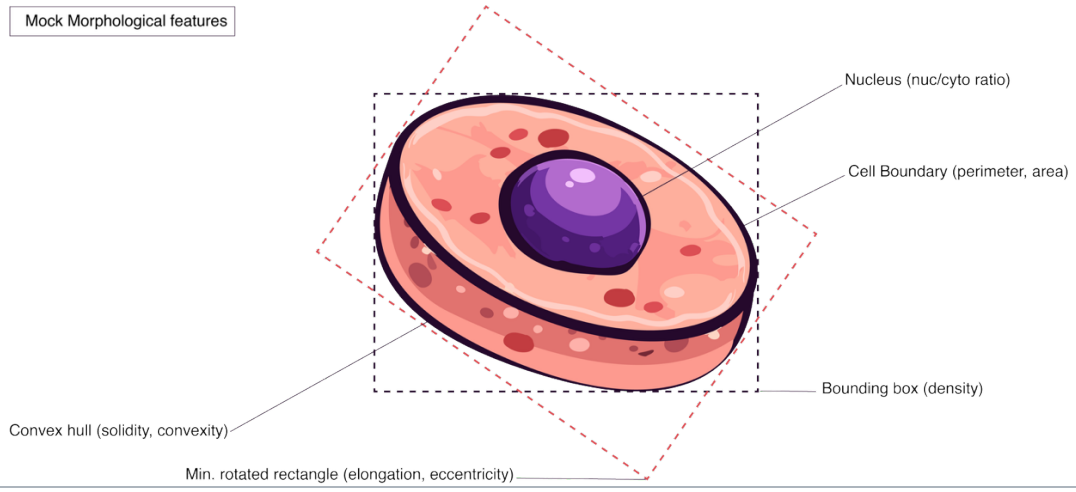
Morphological Metrics:

Each metric below is computed per cell from the polygon boundary geometry using Shapely. Distributions are displayed as violin + box plots across methods. The table lists every formula, its expected range, and what deviations indicate. Many of these metrics are taken directly from the R package CellSPA and we recommend reading their API for more information.

Metric	Formula	Range	Interpretation
cell_area	polygon.area	Tissue-dependent	Extreme outliers suggest over/under-segmentation. Compare against known cell diameters for your tissue type.
elongation	major_axis / minor_axis	≥1 (1 = circular; >3 elongated)	Neurons and fibroblasts are naturally elongated. Round cell types (lymphocytes) should cluster near 1.
circularity	4π · area / perimeter²	0–1 (1 = perfect circle)	Low values indicate spiky or irregular boundaries, possibly from noisy / pixelated detection.
compactness	sqrt(4π · area) / perimeter	0–1 (1 = perfect circle)	Like circularity but less sensitive to perimeter noise. Prefer this when comparing methods with different boundary smoothness.
eccentricity	sqrt(1 − (minor/major)²)	0–1 (0 = circle, ~1 = line)	Bounded complement to elongation. Useful for flagging extremely stretched segments.
solidity	area / convex_hull_area	0–1 (1 = fully convex)	Low solidity means deep boundary indentations — can indicate star like cell morphology or segmentation artifacts.
convexity	hull_perimeter / perimeter	0–1 (1 = fully convex)	Low values flag wavy or fractal-like boundaries, often a sign of noisy pixelated segmentation.
density	area / (bbox_w × bbox_h)	0–1 (1 = fills bbox)	Very low density suggests large empty regions in the bounding box — typical of elongated or curved shapes.
nuclear_ratio	nucleus_area / cell_area	0–1 (typical 0.1–0.5)	Very high (>0.8) = boundary too tight (just the nucleus) typical in lymphocytes. Very low (<0.05) = over-expanded boundary. NA if no nucleus matched.

Visual Reference:

The diagram below illustrates how morphological features are derived from a single cell's polygon boundary. Each geometric construct (convex hull, bounding box, minimum rotated rectangle, nucleus overlay) maps directly to one or more metrics in the table above.



Practical Tips for Choosing a Segmentation

1. Nothing beats the human eye!
2. No single metric is sufficient, compare the full profile across methods.
3. If two methods have similar transcript capture, prefer the one with tighter morphological distributions (lower variance in circularity, solidity, etc.).
4. High '% Cells Without Nucleus' combined with many small cells suggests over-segmentation of cytoplasmic fragments.
5. For tissues with diverse cell types (e.g. brain), expect wider elongation and circularity distributions, this is biological signal, not an artifact.
6. Use the counts-vs-genes scatter to spot systematic doublets (upper-right outliers) and empty segments (lower-left cluster near the origin).